

Predicting if passengers will survive the titanic or not

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1. Introduction

In this project I used the **Titanic dataset** because I wanted to see how we would predict which passenger survived and what the common traits of a survivor of the Titanic are.

2. Dataset Description

The dataset, which contains 890 Titanic passengers' info. Each row includes their age, sex, class, and fare.

3. Cleaning Summary

The raw dataset had some problems that needed fixing before analysis:

- **Missing values:** I found two columns that had some missing values, one of them, the "Cabin" column was mostly missing, so I completely dropped that column. The other one, "Age" had fewer missing values, it had approximately 20% missing, and I dropped them because its hard to predict and fill the ages of passengers.

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4. Models used

I tried three models in this project, LogisticRegression, RandomForest, and KNN, and then I compared their f1 scores to see the best model, and these were the results

- 1-RandomForest(0.61)
- 2-LogisticRegression(0.58)
- 3-KNN(0.46)

5. Neural Network

After that I made some neural networks using ReLU and also softmax / sigmoid, there were 3 dense layers in each, and also I added some dropout layers. after 10 epochs I tested the difference in accuracy, and i found that with the dropout layers accuracy would drop and be less than without the dropout layers

6. Results and Comparison

After training the models and making the neural networks, I compared their F1 score to check which way is best, and I found some surprising results! And the results are as follows:

- 1 - RandomForest (0.61)
- 2 - Logistic Regression (0.58)
- 3 - Sigmoid + Dropout (0.53)
- 4 - KNN (0.46)
- 5 - Softmax + Dropout (0.25)

7. What I learned

I learned that neural networks are not always the best to use for predictions.

8. What I Would Do Next

For future work, I will try to fill the "Age" column, and also I would try more models to see if there is a better one, and also I would try to put more layers in the neural networks and maybe find the optimal number of dropout layers.
